RayTracing

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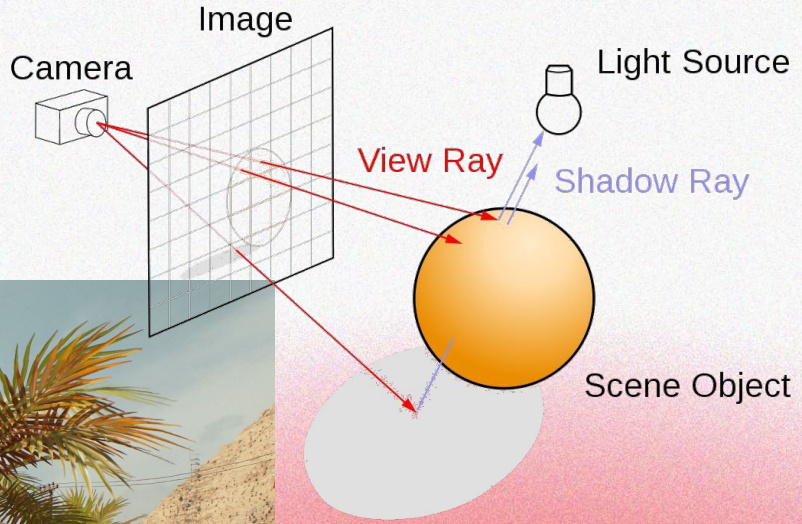
Full Ray Tracing Off

Full Ray Tracing On

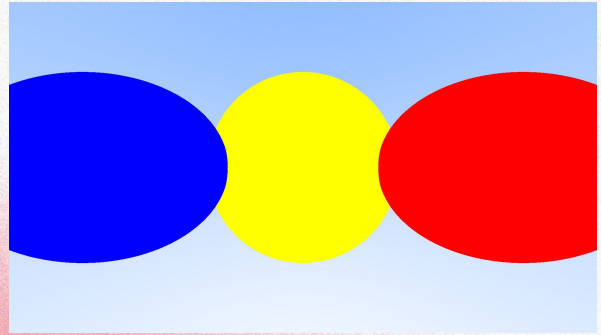


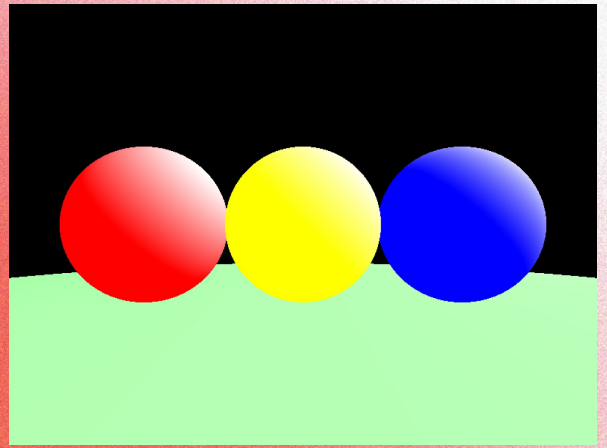
Ray Tracing in One Weekend

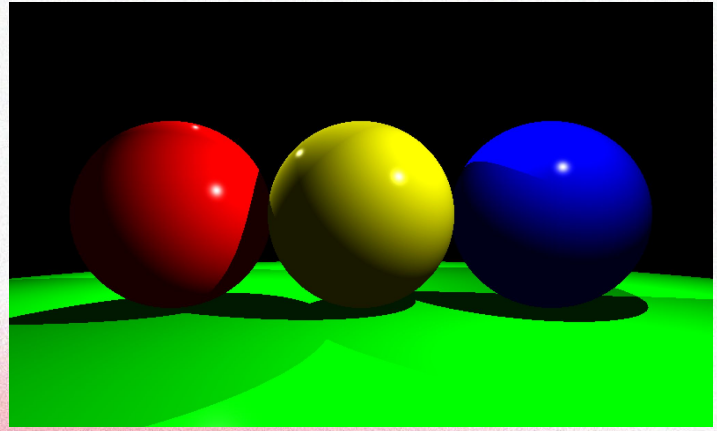
—

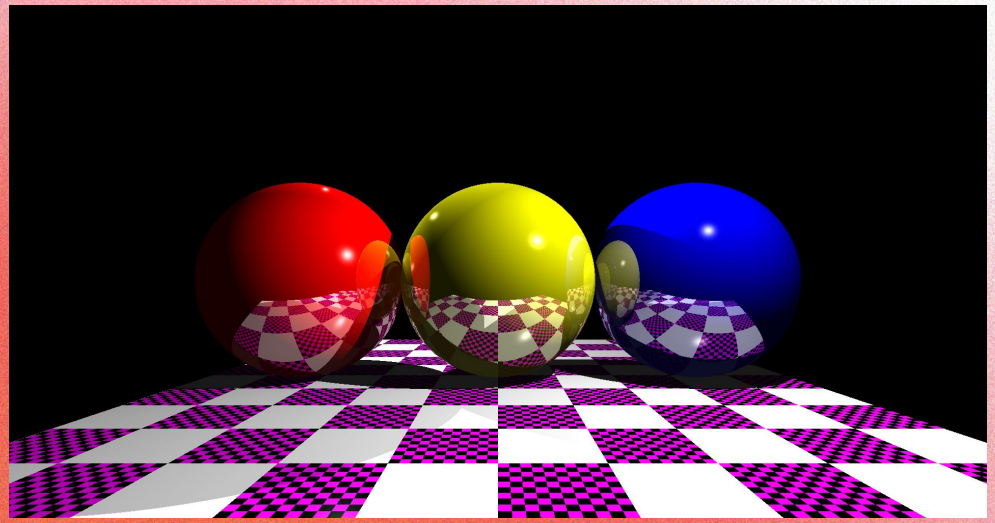
The Book Series

Progreso









Código y paralelización

>> CUDA

>> OOB → AoS

>> No BVH

```cpp
void Renderer::RenderCPU()
{
    for (int y = 0; y < _film->GetTamY(); ++y)
    {
        for (int x = 0; x < _film->GetTamX(); ++x)
        {
            const Ray ray_primary = _camera->GetRay(x, y);
            const Color c = RayColor(ray_primary, k: 10);
            _film->AddPixel(x, y, c);
        }
    }
}
```

```cpp
void Film::AddPixel(int x, int y, Color color)
{
    int idx = (y * _tamX + x) * 4;

    _pixels[idx + 0] = UnitTo255(color.x);
    _pixels[idx + 1] = UnitTo255(color.y);
    _pixels[idx + 2] = UnitTo255(color.z);
    _pixels[idx + 3] = 255;
}
```

```cpp
__global__ void renderKernel( // parametros
{
    const int x = blockIdx.x * blockDim.x + threadIdx.x;
    const int y = blockIdx.y * blockDim.y + threadIdx.y;

    if (x >= width || y >= height || x < 0 || y < 0)
        return;

    const vec3 sample = camera->positionTopLeft
        + camera->deltaX * static_cast<float>(x)
        + camera->deltaY * static_cast<float>(y);
    const vec3 direction = glm::normalize( x: sample - camera->position);
    const vec3 color = rayColor(
        shapes, shapeCount,
        materials, textures,
        lights, lightCount,
        origin: camera->position,
        direction);

    const int idx = (y * width + x) * 4;
    pixels[idx + 0] = unitTo255(color.x);
    pixels[idx + 1] = unitTo255(color.y);
    pixels[idx + 2] = unitTo255(color.z);
    pixels[idx + 3] = 255;
}
```

Métricas

Plataforma: Release x64

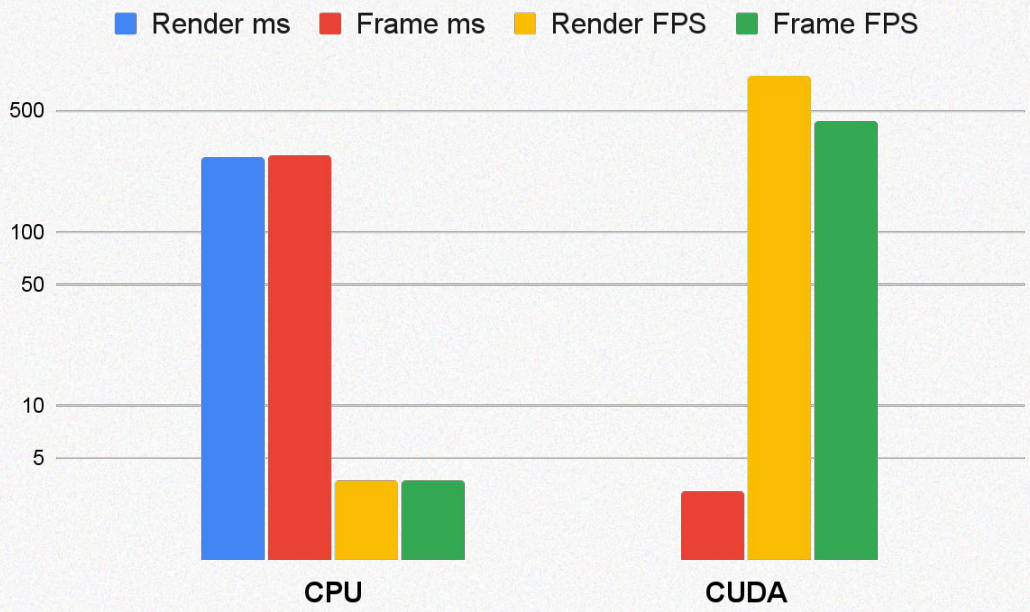
CPU:
Intel i7-12700KF @ 3.60 GHz
(Sólo un core en uso)

GPU:
NVIDIA GeForce RTX 4070 Ti

Conversión de OOP a AoS plano una única vez al principio de la ejecución, imágen estática

¡Un solo sample!

No es coalescente…



Resolución	Speedup
960×540	177.45x
1920×1080	213.28x
7680×4320	252.66x

Bibliografía y referencias

Shirley, P., Black, T. D., & Hollasch, S. 2025. *Ray Tracing in One Weekend: The book series*. https://raytracing.github.io/

Allen, R. 2018. *Ray Tracing in One Weekend in CUDA*. https://github.com/rogerallen/raytracinginoneweekendincuda

Uguray, A. 2026. *Chapters on GPU ray tracing.* https://github.com/RayTracing/gpu-tracing

Gracias